



CS111L – Programming Fundamentals

LAB # 02

Basic Syntax, Declaration and Error

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Task 1: Simple Input and Output

Write a program to input your name and age, then display them.

Code:

```
main.cpp

1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     string name;
7     int age;
8
9     cout << "Enter your name: ";
10    cin >> name;
11
12    cout << "Enter your age: ";
13    cin >> age;
14
15    cout << "Hello " << name << ", you are " << age << " years old." << endl;
16
17    return 0;
18 }
```

Output:

Output

```
Enter your name: Athar
Enter your age: 19
Hello Athar, you are 19 years old.
```

Task 2: Arithmetic Input/Output

Write a program to input two integers and display their sum, difference, product, and quotient.

Code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int a, b;
6
7     cout << "Enter a number: ";
8     cin >> a;
9
10    cout << "Enter another number: ";
11    cin >> b;
12
13    cout << "Sum = " << a + b << endl;
14    cout << "Difference = " << a - b << endl;
15    cout << "Product = " << a * b << endl;
16    cout << "Quotient = " << a / b << endl;
17
18    return 0;
19 }
```

Output:

```
Enter a number: 34
Enter another number: 34234
Sum = 34268
Difference = -34200
Product = 1163956
Quotient = 0
```

Task 3: Average of Three Numbers

Input three floating-point numbers and calculate their average.

Code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float a, b, c;
6
7     cout << "Enter three numbers: ";
8     cin >> a >> b >> c;
9
10    float average = (a + b + c) / 3;
11
12    cout << "Average = " << average << endl;
13
14    return 0;
15 }
```

Output:

```
Enter three numbers: 352
523
999
Average = 624.667
```

Task 4: Student Information

Write a program that asks for a student's roll number, name, and marks in 3 subjects. Display total and percentage.

Code:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main(){
6     int roll, m1, m2, m3;
7     string name;
8
9     cout << "Enter roll number: ";
10    cin >> roll;
11
12    cout << "Enter name: ";
13    cin >> name;
14
15    cout << "Enter marks in 3 subjects: ";
16    cin >> m1;
17    cin >> m2;
18    cin >> m3;
19
20    int total = m1 + m2 + m3;
21    float percentage = (total / 300.0) * 100;
22
23    cout << "\n--- Student Report ---" << endl;
24    cout << "Roll: " << roll << endl;
25    cout << "Name: " << name << endl;
26    cout << "Total Marks: " << total << endl;
27    cout << "Percentage: " << percentage << "%" << endl;
28
29    return 0;
30 }
```

Output:

```
Enter roll number: 2604277
Enter name: Athar
Enter marks in 3 subjects: 99
334
234

--- Student Report ---
Roll: 2604277
Name: Athar
Total Marks: 667
Percentage: 222.333%
```

Salary Slip

Code:

```

1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     float basicPay, houseRent, dailyAllowance;
7     float taxRate, grossSalary, taxDeduction, netPay;
8
9     cout << "Enter Basic Pay: ";
10    cin >> basicPay;
11
12    cout << "Enter House Rent: ";
13    cin >> houseRent;
14
15    cout << "Enter Daily Allowance: ";
16    cin >> dailyAllowance;
17
18    cout << "Enter Tax Rate (%): ";
19    cin >> taxRate;
20
21    grossSalary = basicPay + houseRent + dailyAllowance;    // gross salary
22
23    taxDeduction = grossSalary * (taxRate / 100);    // tax deduction
24
25
26    netPay = grossSalary - taxDeduction;    // net pay
27
28    cout << "\n=== SALARY SLIP ===" << endl;
29    cout << "Basic Pay      : " << basicPay << endl;
30    cout << "House Rent      : " << houseRent << endl;
31    cout << "Daily Allowance : " << dailyAllowance << endl;
32    cout << "Gross Salary   : " << grossSalary << endl;
33    cout << "Tax Rate       : " << taxRate << "%" << endl;
34    cout << "Tax Deduction  : " << taxDeduction << endl;
35    cout << "Net Pay        : " << netPay << endl;
36    cout << "===== " << endl;
37
38    return 0;
39 }

```

Output:

```

Enter Basic Pay: 100000
Enter House Rent: 12000
Enter Daily Allowance: 3203

Enter Tax Rate (%): 77

=== SALARY SLIP ===
Basic Pay      : 100000
House Rent     : 12000
Daily Allowance : 3203
Gross Salary   : 115203
Tax Rate       : 77%
Tax Deduction  : 88706.3
Net Pay        : 26496.7
=====

```

Ahhh.. so much tax!